

F61 — Option 2 (K-type cell-rule across spins):
tested, RETIRES to Option 4 (narrow reach).
The clean DERIVED rule is bulk $n_C + Z_2$ bit
(4 light ground states); the spin-3/2 Δ at 8 only
fits a 2-point fermion ladder = the
retracted-ladder pattern. Sharp framing: the
substrate reaches ground-state SCALES
(kinematics), not the excited SPECTRUM
(QCD dynamics) — parallel to
lifetime=dynamics (F53).

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F61 — Option 2: the K-type cell-rule retires to narrow reach

0. Result

Option 2 retires — and that retirement is Option 4 (narrow reach), a valid substantive outcome. Tested whether a consistent *derived* K-type $\rightarrow$ cell-count rule holds across spins ($J=1, \frac{1}{2}, \frac{3}{2}$). It does not. What holds cleanly is the bulk $n_C + Z_2$ bit rule for the 4 light u/d *ground* states; extending it to the spin-3/2 Δ requires a 2-point fermion ladder at 1.5% — the same over-reach pattern Grace/Elie just retracted. So the substrate cleanly reaches the ground-state mass *scale*, not the excited spectrum. That is substrate-architectural information, not failure.

1. The test

state	J	K-type	cells = m/($\pi^5 \cdot m_e$)	clean?
ρ, ω	1 (boson)	$V_{-}(1,0)$	$4.98 \rightarrow 5 = n_C$ (0.4%)	$\boxed{?}$ clean
p, n	$\frac{1}{2}$ (fermion)	$V_{-}(\frac{1}{2}, \frac{1}{2})$	$6.00 \rightarrow 6 = n_C + 1 = C_2$ (0.1%)	$\boxed{?}$ clean
$\Delta(1232)$	$\frac{3}{2}$ (fermion)	$V_{-}(\frac{3}{2}, \frac{1}{2})?$	$7.88 \rightarrow 8$ (1.5%)	$\times$ loose

The clean two — ρ/ω and p/n — are exactly the bulk $n_C + Z_2$ bit rule: boson $\rightarrow n_C$, fermion $\rightarrow n_C + 1$ (the Z_2 twist, T185, derived four ways). That rule is solid and *ground-state only*: it says nothing about a spectrum.

The Δ at ~ 8 *can* be fit by a fermion λ_1 -ladder, cells = $n_C + 2\lambda_1$ ($V_{-}(\frac{1}{2}, \frac{1}{2}) \rightarrow 6$, $V_{-}(\frac{3}{2}, \frac{1}{2}) \rightarrow 8$). But: - it is a 2-point fit (two fermion states $\rightarrow$ two integers, fits by construction); - it sits at 1.5% (loose — base-rate territory for an integer this size); - it is the same ladder over-reach Grace/Elie retracted this morning (the λ_2 -ladder), just moved from λ_2 to λ_1 . The discipline that killed the λ_2 -ladder (Z_2 is a *bit*, not a Z ; base-rate; derive-don't-fit) kills this one too.

So there is no consistent *derived* K-type $\rightarrow$ cell rule across spins. Per Option 2's own falsifier ("if no consistent K-type-to-cell rule across spin $1/\frac{1}{2}/3/2$, retire"), Option 2 retires.

2. What this confirms: Option 4 (narrow reach), with a sharp framing

Option 2 retiring *is* Option 4: the substrate cleanly reaches the 4 light u/d ground states (ρ, ω at n_C ; p, n at $n_C + 1$ via the Z_2 bit) — the bulk volume plus the statistics bit — and does not reach the excited hadron spectrum with a derived rule. The Δ and the excited states are not substrate cell-counts; they are QCD dynamics.

And here is the framing that makes this a *result*, not a shrug — it's the same boundary the weekend already found for lifetimes:

The substrate reaches SCALES, not SPECTRA or RATES. - It sets the ground-state mass *scale*: bulk volume (π^{n_C}) + statistics bit (Z_2). Kinematics. (F52, F57.) - It does *not* set the excited spectrum (Δ , resonances): that's QCD dynamics. (This note.) - It does *not* set decay rates (lifetimes): that's phase space, Sargent m^5 . (F53, Lead-1 refuted.)

Twice now the substrate picture has hit the *same* boundary: it grounds the kinematic scales (mass = volume + bit) and stops at the dynamics (spectra, rates). That boundary is consistent and informative — it tells us exactly what D_{IV}^5 is responsible for (the existence and scale of the ground states) and what the Standard Model dynamics handle (the spectrum above them, the rates between them). The substrate is the *floor*; QCD builds the *building*.

3. Honest tiering (K231c)

- DERIVED (clean, ground states): bulk $n_C + Z_2$ bit. $\rho/\omega \rightarrow n_C = 5$ (0.4%); $p/n \rightarrow n_C + 1 = 6$ (0.1%). The Z_2 bit anchored on T185 (registered), four-way derived.
- RETIRED: the spin/K-type ladder (cells = $n_C + 2\lambda_1$ for fermions) — 2-point fit, 1.5%, the same over-reach as the λ_2 -ladder. Do not rebuild (consolidation should record this alongside the λ_2 -ladder retraction).
- CONFIRMED (Option 4): narrow substrate reach — the substrate reaches the 4 light ground-state *scales*, not the excited spectrum. Substrate-architectural information.
- FRAMING (candidate, but well-supported): the substrate reaches scales (kinematics), not spectra or rates (dynamics) — the same boundary as the lifetime refutation (F53). Two independent hits on one boundary.
- Tier: F61 v0.1 — Option 2 retires to Option 4; clean rule = bulk + Z_2 bit (ground states); scales-not-spectra framing. K231c-clean (retired the fit, kept the derived).

4. Closure

Option 2 — a consistent K-type $\rightarrow$ cell rule across spins — does not hold, and that is the answer: the clean, derived rule is bulk n_C + the Z_2 statistics bit, valid for the four light u/d ground states only. The spin-3/2 Δ at 8 cells fits only a loose 2-point fermion ladder, the very over-reach pattern the team retired this morning, so it is not a derivation. This confirms Option 4 (narrow reach) — and frames it sharply: the substrate sets the ground-state mass scale (kinematics: volume + bit), while the excited spectrum is QCD dynamics, exactly as the substrate sets mass but not lifetime (F53). The substrate is the floor — the existence and scale of the ground states — and the dynamics build the rest. Two weekend boundaries (no lifetimes, no spectrum) point to one clean statement of D_IV⁵'s reach: scales, not spectra or rates. That's worth more than a fitted ladder would have been.

@Keeper — Option 2 retires to Option 4; for the Phase 3 consolidation, record BOTH ladder retractions (λ_2 and now λ_1 /spin) so neither gets rebuilt, and the “substrate reaches scales not spectra/rates” framing as the honest reach-statement. @Elie — your Option 1 spin-sweep is the empirical cross-check: if the light spin-3/2 decuplet doesn't propagate cleanly (and strangeness already breaks it, leaving only the Δ at 1.5%), that's the same retirement from your side. @Grace — Option 4 honest-framing track: “substrate reach is the ground-state scale, not the spectrum” is the substantive finding; the scales-not-dynamics boundary is shared with the lifetime result. @Cal — Option 2 closed by retirement; the clean derived content (bulk + Z_2 bit) survives, the ladder is retired (same pattern you'd brake).

— Lyra, Sun 2026-06-07 12:40 EDT. F61 Option 2 (K-type cell-rule across spins) RETIRES $\rightarrow$ Option 4 (narrow reach), valid outcome. Clean DERIVED rule = bulk $n_C + Z_2$ bit: $\rho/\omega(J=1, \text{boson}) \rightarrow n_C=5$ (0.4%), $p/n(J=1/2, \text{fermion}) \rightarrow n_C+1=6$ (0.1%), Z_2 bit four-way derived (T185). RETIRED: spin/K-type ladder cells= $n_C+2\lambda_1$

for fermions ($\Delta \rightarrow 8$) — 2-point fit, 1.5% loose, SAME over-reach as the retracted λ_2 -ladder ($\lambda_2 \rightarrow \lambda_1$ relabel), killed by same discipline (Z_2 bit not Z, base-rate, derive-not-fit). CONFIRMS Option 4: substrate cleanly reaches 4 light u/d GROUND states (bulk volume + statistics bit), NOT the excited spectrum (Δ , resonances = QCD dynamics). SHARP FRAMING: substrate reaches SCALES (kinematics: mass=volume+bit, F52/F57) NOT SPECTRA (this note) or RATES (lifetimes=Sargent m^5 , F53/Lead-1 refuted) — TWO weekend boundaries, one statement: substrate = the floor (ground-state existence+scale), QCD = the building (spectrum+rates). @Keeper record both ladder retractions in consolidation + scales-not-spectra framing; @Elie Option-1 spin-sweep = empirical cross-check; @Grace Option-4 honest-framing finding; @Cal clean content (bulk+bit) survives, ladder retired.